

A TRANSFORMER-BASED FRAMEWORK FOR VIDEO ANOMALY DETECTION

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Introduction

The rapid growth of surveillance systems in smart cities has increased the need for efficient and accurate monitoring solutions, as traditional methods relying on human operators are time-consuming and error-prone.

Deep learning enables automatic analysis of video data by learning normal patterns and detecting abnormal events in real time.

This project proposes a Transformer-based model that leverages spatial and temporal features to enhance anomaly detection in real-world surveillance scenarios[1].

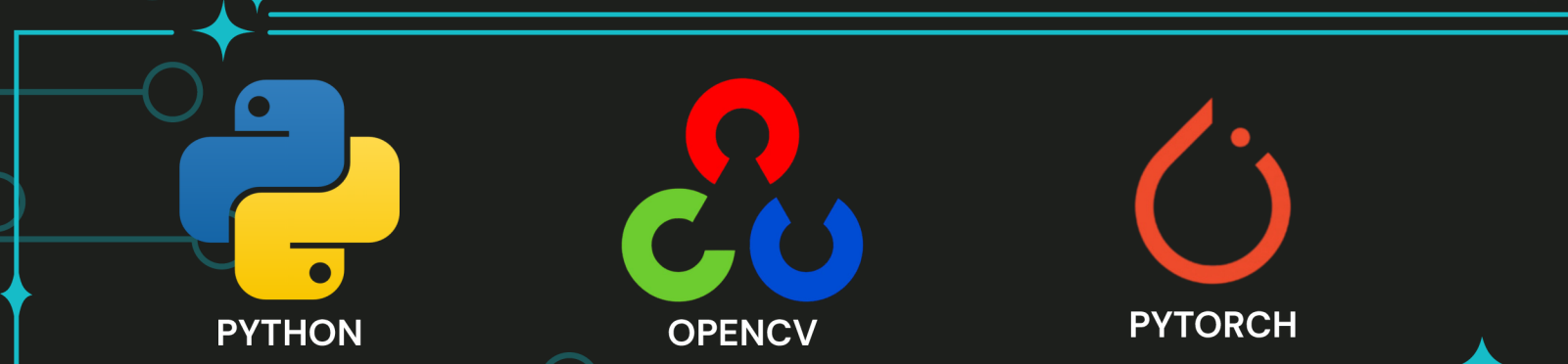
EXAMPLES OF ANOMALIES



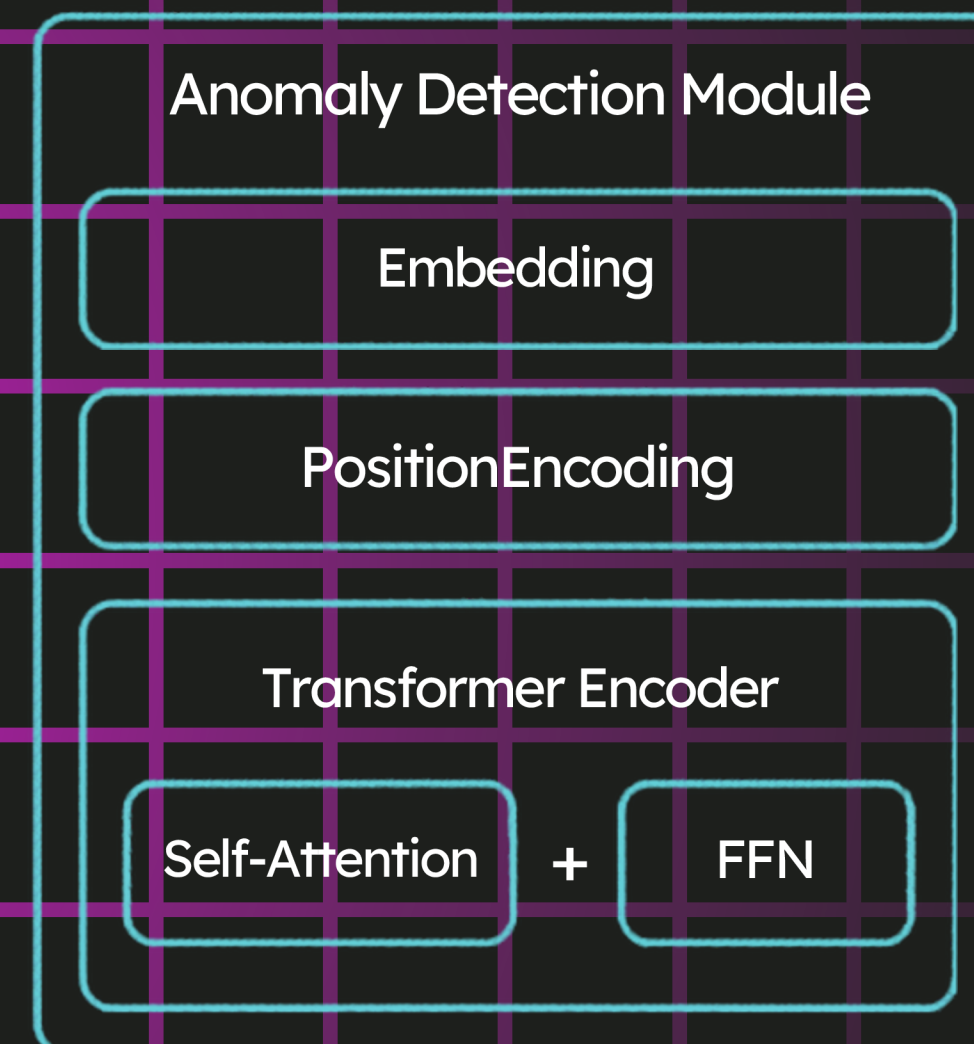
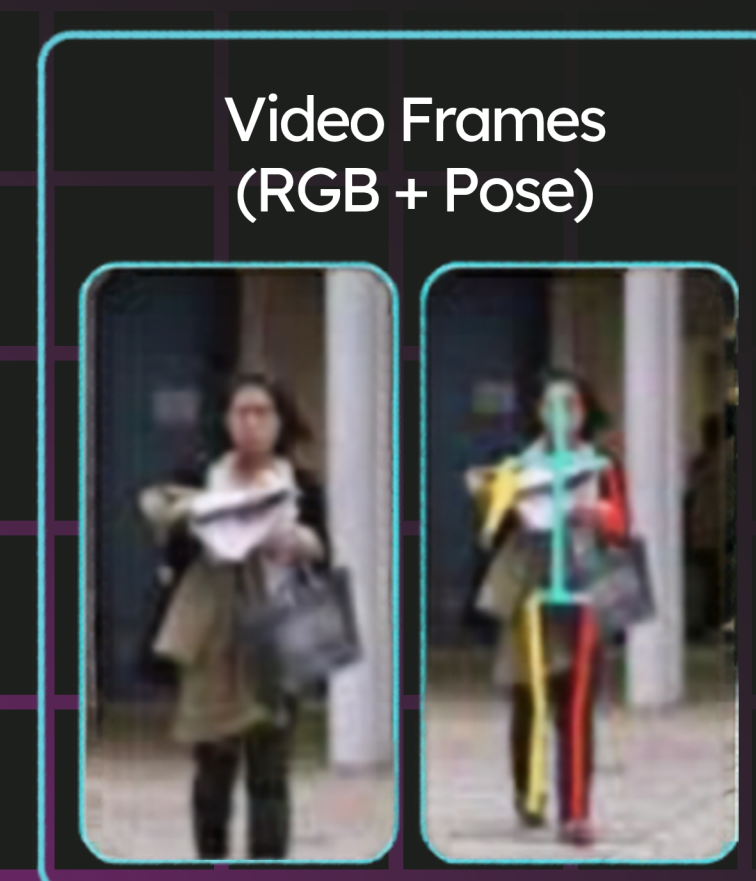
Motivation

- Traditional surveillance systems rely on continuous human monitoring, which is time-consuming, costly, and prone to errors.
- The rapid advancement of deep learning enables accurate detection of abnormal events in video data while reducing manual effort.
- This project supports the development of smart surveillance systems aligned with Saudi Vision 2030 to enhance public safety and build smarter, safer cities.

Tools



Architecture



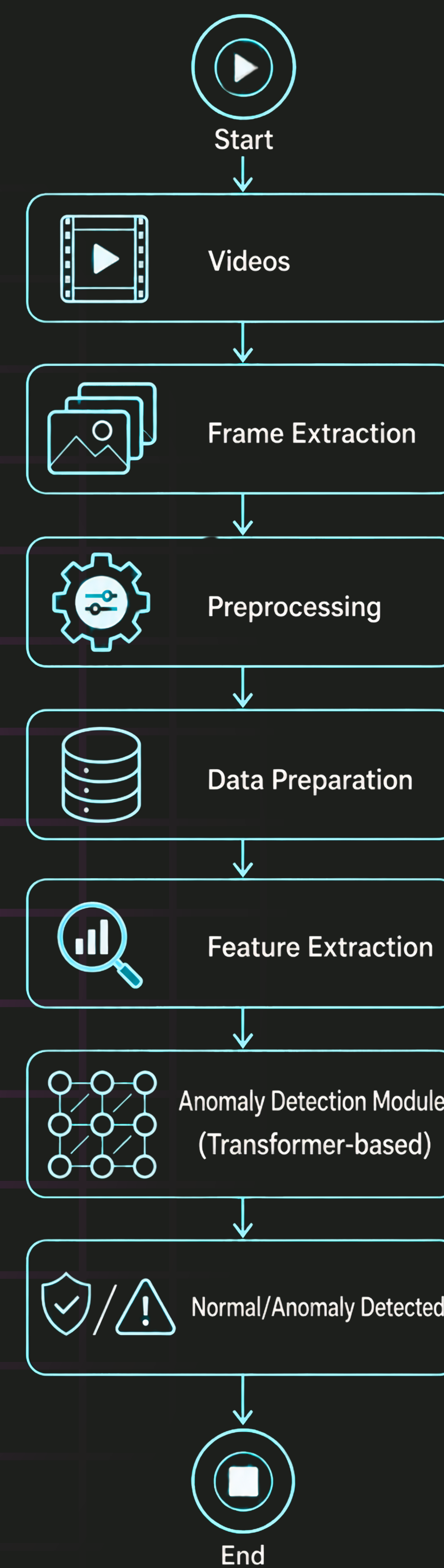
Transformer-based Anomaly Detection Module

Abnormal Detection

Output

Methodology

Pipeline



Experiment

In this experiment, we evaluated the performance of three different input modalities: RGB, Pose, and RGB+Pose (Fusion) for video anomaly detection using the Avenue dataset[2]. The models were compared using the AUC-ROC metric to measure their ability to distinguish between normal and abnormal events.

RGB	POSE	RGB+POSE
81.28%	69.28%	79.04%

Conclusion

The proposed Transformer-based framework highlights the effectiveness of modeling temporal patterns for accurate anomaly detection in videos. It provides a promising step toward scalable and real-world intelligent surveillance systems.



References